Scalar coordinates for retained Jacobian relations 2026-10-31

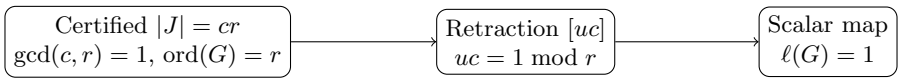
T-1120. Producer derivation/source audit; independent review pending.

Derived criterion

For a finite abelian group J, prime r and $G \in J$, an additive map $\ell : J \to \mathbf{F}_r$ with $\ell(G) = 1$ exists exactly when $G \notin rJ$. Every such map kills rJ. Conversely J/rJ is an $\mathbf{F}_r$-vector space; extend its nonzero vector $\bar{G}$ to a basis and take that coordinate functional. For retained relations, the same criterion on the subgroup K generated by all retained classes and G suffices. Its map need not extend to J.

Explicit sufficient construction

If $|J| = cr$, $\gcd(c, r) = 1$ and $\operatorname{ord}(G) = r$, choose $uc \equiv 1 \pmod{r}$. The quotient $J/\langle G\rangle$ has order c, hence $[c]X \in \langle G\rangle$. Thus $[uc]$ is a retraction fixing G. Define $\ell(X)$ uniquely by $[uc]X = [\ell(X)]G$. This proves existence and additivity, not an efficient coordinate-evaluation algorithm.



Two boundary examples

In $J = \mathbb{Z}/49\mathbb{Z}$, $r = c = 7$, $G = 7$ has order seven. Every X passes $[7]([7]X) = 0$, but every map to $\mathbf{F}_7$ kills G. Even exhaustive torsion checks cannot prove normalization. In contrast, $J = \mathbf{F}_7^2$, $G = (1, 0)$ admits $\ell(x, y) = x$ although its cofactor is seven. Coprimality is sufficient, not necessary. These are abstract derivations, not cohort curve results.

Source binding and next check

At source `a34ce033054a`, `native_solver.rs:1188--1280` checks prime r, cofactor arithmetic, source-generator nonzeroness and r-torsion of its transfer. It checks projected torsion for sixteen rational points; its README already calls this diagnostic. That excerpt has no explicit nonzero-transfer or coprime-cofactor assertion. Lines 1311–1359 verify $D + \sum Q_i = 0$; lines 471–515 store signed coefficients and augmented columns $(b, -a)$.

With additive transfer, $H = [k]P$, $G = \tau(P)$ and a normalized ℓ, this identity gives $\sum \ell(Q_i) + bk = -a$. The point-class negation identity supplies the signs; order-two classes vanish for odd r. Bind destination cardinality, transfer additivity/nonzeroness and retained point semantics next. The complete report gives exact obligations and hashes. No scientific workload or timing was run; cost ratio remains null.

Sources and status

Standard group facts: Mathlib OrderOfElement (subgroup index and cardinality annihilation), and basis extension. Background: Arita et al., IEICE 2006, p.1252, explicit Scholten pullback example. Original Scholten PDF inaccessible; ePrint endpoint returned 403. No novelty claim. This derivation is not yet kernel-checked or independently reviewed. Review agents remain usage-limited. The expired campaign envelope is unchanged.